

$$(S_T - K)^+ = 130 - 100 = 30$$

$$T \mapsto r(T) \quad DF(T) = e^{-r(T)T}$$

$$e^{r(t)t} e^{r_{\text{fwd}}(T-t)} = e^{r(T)T}$$

$$\sigma \sqrt{T-t}$$

$$\frac{dS_t}{S_t} = (r - d)dt + \sigma dW_t$$

$$S_0 - K e^{-nT}$$

$$\theta = \frac{\partial C_0}{\partial t}$$

$$\Gamma(u(\text{Gain}))$$

$$C_0 - P_0 = S_0 - K e^{-nT}$$

$$\Delta_C - \Delta_P = 1$$

$$C_t - P_t = S_t - K e^{-n(T-t)}$$

$$V_C - V_P = 0$$

$$V_t = K e^{-r(T-t)} - S_t \quad \nearrow \text{quand } t \nearrow$$

$$(K - S_T)^+ \approx K - S_T$$

$$P_0 \approx K e^{-rT} - S_0$$

$$C_0 - P_0 \approx S_0 - K e^{-rT} \quad \approx \quad C_0 \approx 0$$

$$P_0 \approx K e^{-rT} - S_0$$

$$C_t \approx S_t - Ke^{-r(T-t)}$$

$$N(x) \approx \underbrace{N(0)}_{\frac{1}{2}} + \underbrace{N'(0)}_{\frac{1}{\sqrt{2\pi}}} x$$

$$C_0 = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

ATM forward $K = S_0 e^{rT}$

$$d_1 = \frac{1}{2} \sigma\sqrt{T} = -d_2$$

$$C_0 = S_0 \left(N\left(\frac{1}{2} \sigma\sqrt{T}\right) - N\left(-\frac{1}{2} \sigma\sqrt{T}\right) \right) \approx \underbrace{\frac{1}{\sqrt{2\pi}}}_{\approx 0,4} S_0 \sigma\sqrt{T}$$